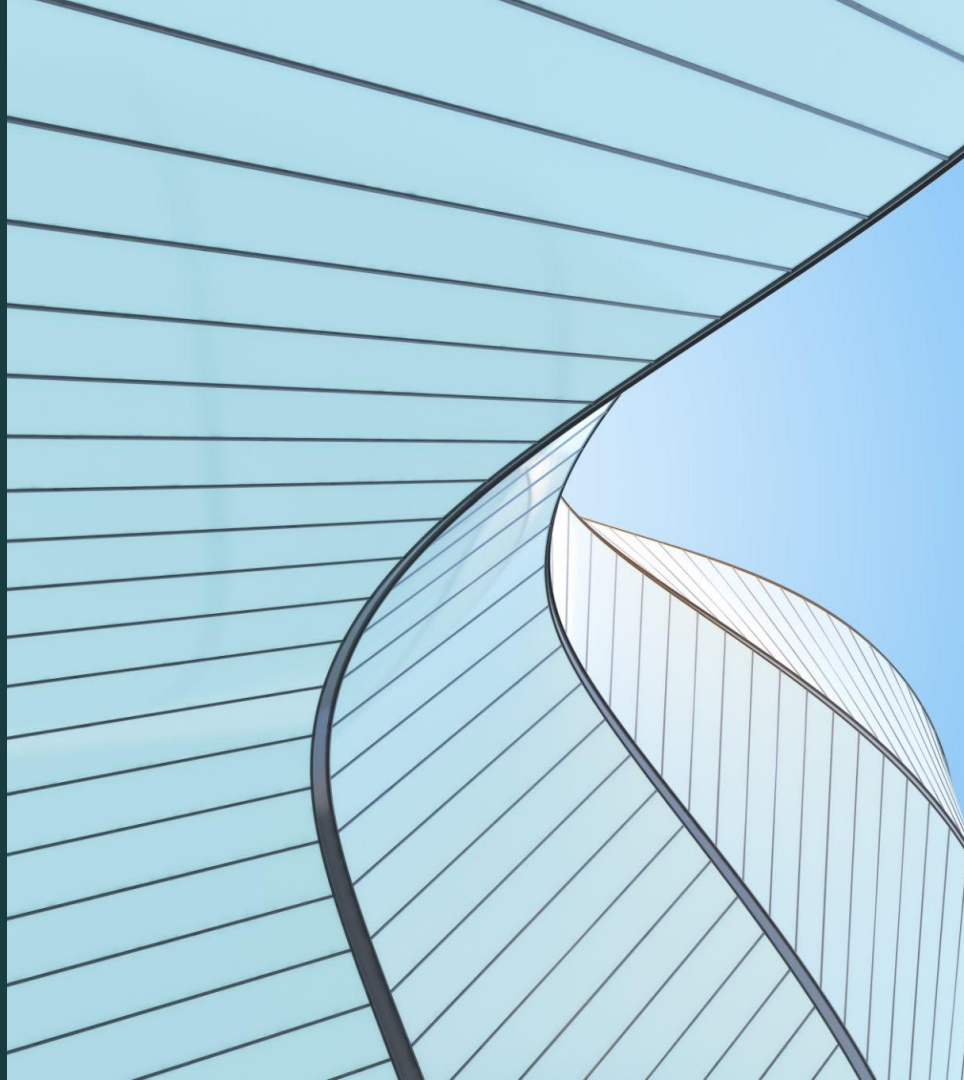
Existing methods and literature

Group D: Arnav K, Alex, Arthur, Emily



Content *overview*

1

What is our project, and what model will we use?

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Image Modifications

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Feature Modifications

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Questions

Project and model we will use

- Our project aims to train a model in prostate segmentation at one site and measure how well it generalises to unseen hospitals (unlike SAML which trained on 5 and generalised to 1)
- We will use the same sites as in SAML – 6 sites comprised of 3 datasets: NCI-SCBI (A & B); I2CVB (C); PROMISE12 (D, E, and F)
- The model we will use is the nnU-Net v2 2D segmentation model

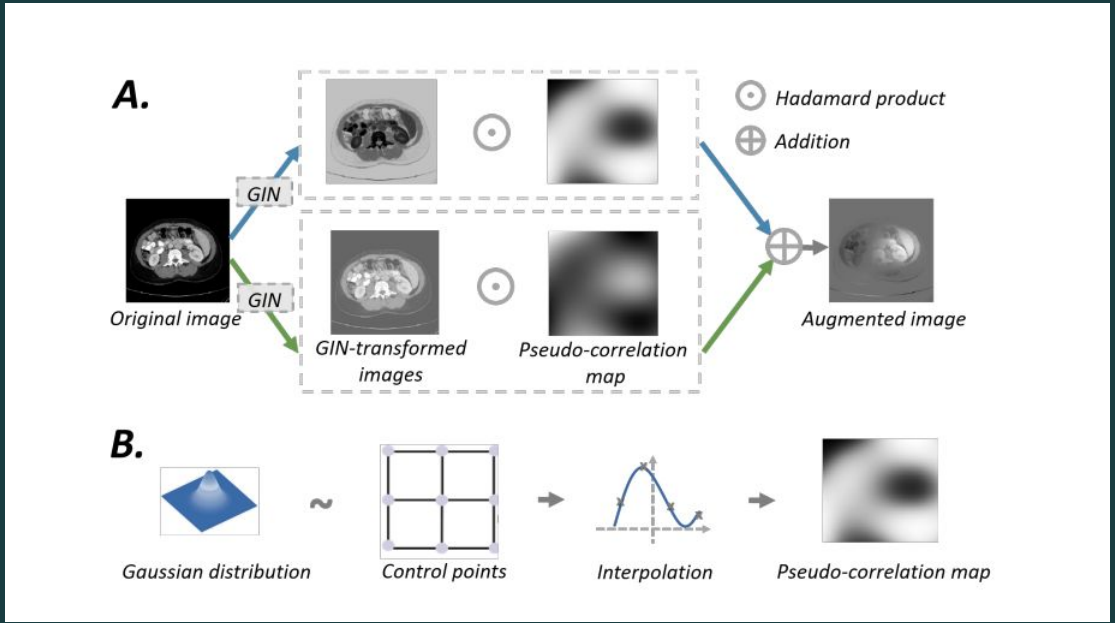
Dataset	Institution	Case num	Field strength(T)	Resolution(in/ through plane)(mm)	Endorectal Coil	Manufactor
Site A	RUNMC	30	3	0.6-0.625/3.6-4	Surface	Siemens
Site B	BMC	30	1.5	0.4/3	Endorectal	Philips
Site C	HCRUDB	19	3	0.67-0.79/1.25	No	Siemens
Site D	UCL	13	1.5 and 3	0.325-0.625/3-3.6	No	Siemens
Site E	BIDMC	12	3	0.25/2.2-3	Endorectal	GE
Site F	HK	12	1.5	0.625/3.6	Endorectal	Siemens

Image augmentation: Causality Inspired SSDG

Setup:
U-Net (EfficientNet-b2 encoder).
Each SAML site used as the source in turn; Dice averaged over the 5 targets.

Key Advantage:
If appearance is random, the network has to learn shape, which does not change between hospitals.

Limitation: Randomness is unconstrained: not tied to how MRI scanners actually differ.



Prostate Dice (%) on unseen sites: plain training vs. GIN+IPA

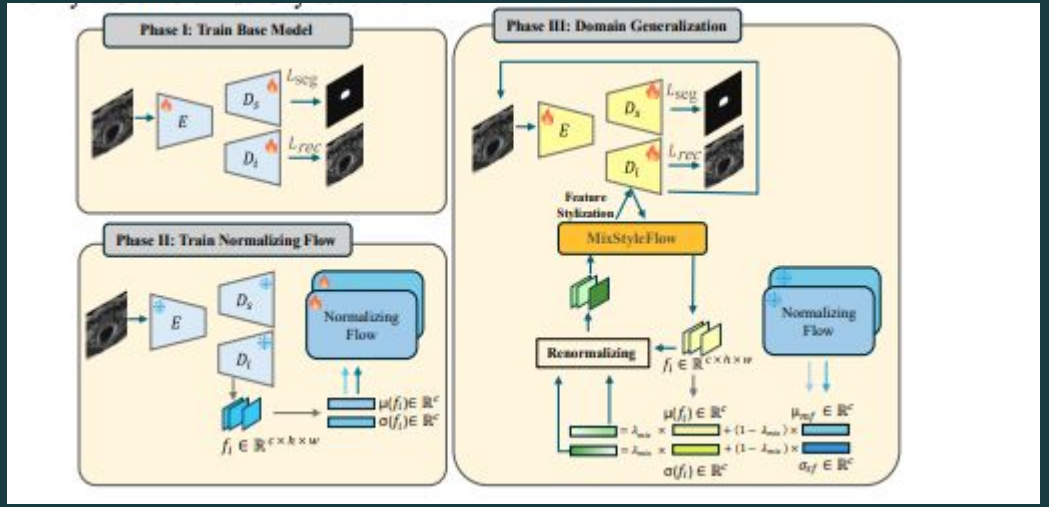
56.6 → 70.4

Feature Augmentation: MixStyleFlow

Setup:
Trained on Medical Decathlon prostate; tested on all six SAML sites.

Key Advantage:
Learns the full distribution of styles with a normalizing flow, rather than just mixing existing examples.

Limitation: Different Training Data



Prostate Dice (%) on unseen sites:

70.5 → 81.1

BIDMC (+18) and HK (+20).

Target Results: Lipschitz regularization

Setup:
Same as us (trained on Site A, testing on others)

Key Advantage:
Penalize the encoder when its output shifts too much with high-frequency detail (scanner texture), and when it ignores mid-frequency detail (shape).

Limitation: Gains are uneven

Table 1. Single-source domain generalization performances in terms of Dice scores for a variety of models with and without Lipschitz regularization.

	(source)	(targets)					Avg on
	RUNMC	BIDMC	BMC	HK	UCL	HCRUDB	targets
UNet	88.09	47.19	78.34	77.98	82.40	80.20	73.22
+Lipschitz	88.51	60.51	80.44	85.85	81.56	79.75	**77.62**
ResUNet	86.97	56.12	79.03	82.75	78.61	77.57	74.82
+Lipschitz	88.61	62.94	80.32	83.62	79.86	80.74	**77.51**
ResUNet++	89.17	60.71	69.95	86.85	78.20	66.87	72.51
+Lipschitz	88.22	64.52	73.31	85.93	77.33	78.74	**75.97**
BayeSeg	85.0	71.2	82.6	82.0	79.7	73.5	77.8
+Lipschitz	87.4	76.9	83.3	84.8	82.2	82.5	**81.94**

Questions?

Resources

https://arxiv.org/pdf/2111.12525

https://papers.miccai.org/miccai-2025/paper/3460_paper.pdf

https://link.springer.com/chapter/10.1007/978-3-031-72117-5_62